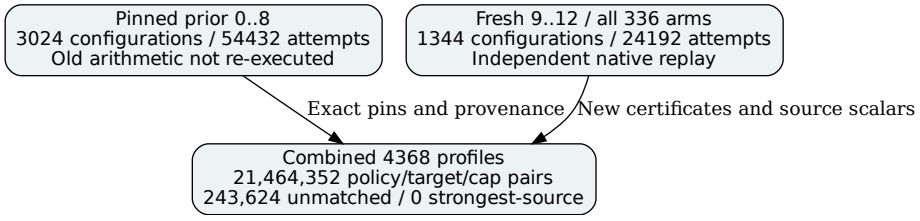
F37 size-three relations above the cap-8 boundary

2026-10-30 PR2381 exact finite native computation

The actual fixed route is $E/\mathbf{F}_{37} : y^2 = x^3 + 7x + 19$ to its nonsquare $d = 35$ twist $T : y^2 = x^3 + 28x + 33$. The existing coordinate map sends $P = (1, 8)$ to $U = (32, 29)$, order 19, with inverse $[10]D$ on the cyclic subgroups. Prior actual-source and cubic-input Lean receipts are reused within their hypotheses, not presented as new or combined audits. No alias is fabricated.

The earlier size-three run completed caps 0–8, but every recovery was at cap 8. This new protocol tests all 336 arms at fresh caps 9–12, keeping the old records immutable and separately labelled. Both use all 84 canonical policies per curve plus all matched/pullback controls and one original encoding.



Fresh upper-range outcome	Count
New configurations	1344
New separate original-source recovery attempts	24192
New verified original-source recovery runs	24192
New blocked public-log setups	0
New capped diagnostic query records	1088
Genuinely changed controls rejected101/no receipt	11
Shared backend tests passed	22

At every upper cap, each role has 1,512 verified recovery runs. At cap 9, the source and pullback roles certify 1,242 diagnostic queries; the twist and matched roles certify 1,238. At caps 10–12, each role certifies all 1,512 queries. All recoveries verify the scalar on ORIGINAL E/P. These counts repeat eighteen source points across policies and budgets; they are not 24,192 distinct DLP challenges.

The combined 4,368 profiles include 3,024 old and 1,344 new configurations. Native analysis evaluates 21,464,352 full policy/target/cap comparisons across 0–12: 243,624 unmatched strict higher-rank/lower-requested-cap recovery-admitted pair records, ZERO against the strongest canonical source frontier. These are pair records, not unique wins. The earlier boundary is resolved within this range; no general no-advantage or practical speedup conclusion follows.

What the expanded budget establishes

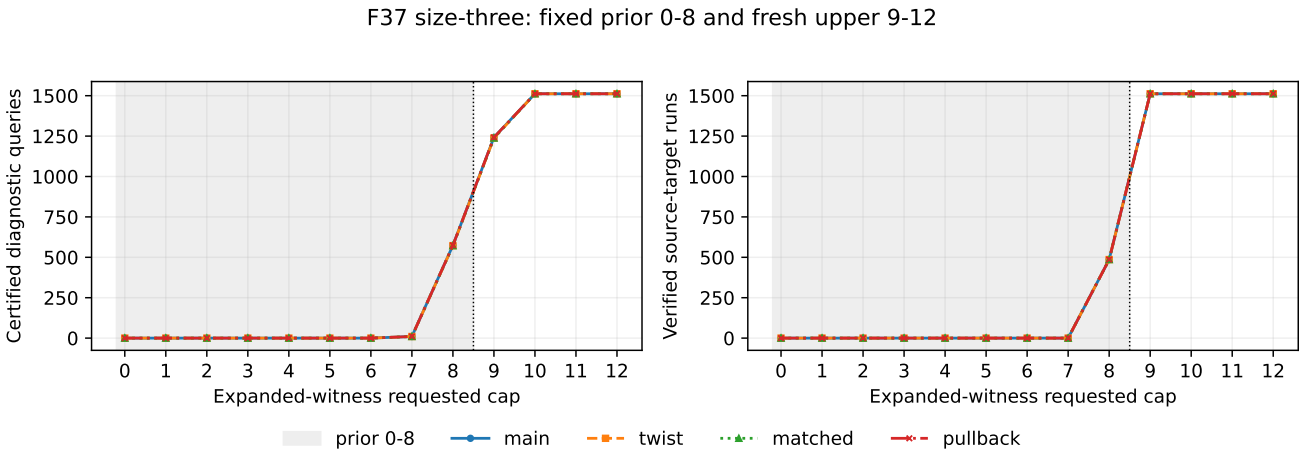


Figure. Native-emitted counts only, 84 policies and 18 source points per role/cap. Grey denotes fixed earlier data, white fresh executions. Curves may overlap. No clock, cross-budget speed ratio or sampling interval is inferred.

The new producer really executes only caps 9–12. The new independent auditor recomputes all 1,344 upper cases with separate field/group/polynomial/linear arithmetic. Old 0–8 records are imported by exact index, audit, protocol, source and case hashes; their arithmetic is not re-executed or counted as fresh. Query/setup/recovery source bytes are unchanged. Combined profiles retain explicit origin labels; no compatibility assumption from filenames alone.

The solver still receives S_3 and both cubic membership equations. Matrix and expanded original-input verification-witness degree fit each rung before constructing shifts. Complete unit/GB certification precedes extraction; unknown caps withhold all rows. The candidate never stops from exhaustive reference roots. Identical-polynomial witness retention stays local to one solve. No free cache or unbounded fallback was introduced.

Setup derives logs only from admitted actual relations until rank three and point-checks each base. Unknown Q alone enters recovery; all failed and identity shifts and ordered sign lifts remain. Fresh solves precede diagnostic output binding. Source scalars are verified on original E/P. A separate direct point reference has no named matrix degree. Setup, arithmetic, allocation and verification lack a calibrated total price.

Higher requested budgets remove setup blocks here, but do not establish a lower actual solving degree, stationary useful yield or intrinsic regularity. The strongest-source joint count stays zero after eligible comparisons exist. Complete pricing, matched strong rho and fastest external comparisons are still required before a speed claim.

Evidence, failures and the next encoding

All fresh cases, altered payloads, inputs, sources, commands, exits and receipts are retained. Ten value/coverage controls plus a rehashed false lower-reference control reject 101 without receipts. The missing-failed-query control is real: 1,088 fresh diagnostic failures exist. The implemented explicit-null rule was not used. The same 22 backend tests pass; no new root-suite pass is claimed.

Canonical protocol/source hashes were frozen before new data. A copied ledger level still says original/reduced and its source-binding prose is too broad; a dated erratum explains these metadata errors without thawing the record. The question, primary metric $336\times1\times4$, arm parameters, Markdown, source loops and coverage assertion already specify one encoding and typed old/new provenance. No criterion or population changed; no blinded statistical claim. Schema checks alone do not approve every prose field.

The first ordinary backlog-helper build missed link dependencies; the official unmodified CLI then compiled with the canonical SHA1/SHA256 modules. An initial card paragraph failed the native length rule and was corrected without changing the rule. Exact failures and successful manifests/lock/source bindings remain. No coordinator was invented: card status stays draft, with no main-ancestry, Cairn lease, funding or task promotion. No new Lean theorem or field identity is claimed; prior19/22,6/7 and eighteen identity records stay unchanged.

Next step, not executed. Prove the symmetric-coordinate cubic-membership split before a new solver experiment: distinct-root and diagonal branches, coverage, lifting, denominator and duplicate controls. The known symmetric S_3 expression has lower unweighted degree, but that does not imply a lower original-coordinate or weighted witness degree, regularity, or cost. Keep these measures separate and compare the entire source-policy frontier. No new branch proof or encoding run is claimed here.

Same PR2381 stays open/non-draft, auto-merge off. Missing 1,921 registrations, current required CI/review, Conductor 127/no lease, historical incomplete library and naming, geometry/probability/regularity/full-cost gates remain. Scientific window 16:36–17:36UTC used two existing local workers and 600-second stage caps. No paid resource, arbitrary result upload or broad credential reuse.

Adjacent README/validation link the full protocol, old/new origin pins, raw reconstruction, native receipts, primary references, graphs and hashes.